MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER PRACTICAL EXAMINATION NOVEMBER 2024

MG1MDCPHY100 – Physics Around You

Duration: 2 hrs **Maximum Marks: 35**

Record:5 marks

Attempt the question marked 'X'. Change of question is not allowed :30 marks

1	Demonstrate Ohm's Law using the following apparatus: a resistor, ammeter, voltmeter, variable power supply, and connecting wires.	[An,S][5]
2	Find the radius of the wire, volume of the sphere and glass piece by measuring its dimensions using Screw gauge.	[A,S][5]
3	Find the volume of the sphere and cylinder by measuring its dimensions using Vernier calliper.	[A,S][5]
4	Demonstrate the use of a digital multimeter (DMM) to test diodes and measure electrical properties, including current, voltage, resistance, and capacitance.	[An,S][5]
5	Using a Cathode Ray Oscilloscope (CRO), investigate and analyze the waveforms generated by a function generator set to produce sine, square, and triangular waves.	[An,S][5]
6	Demonstrate standing waves using Melde's string experiment. Include the key steps in setting up the apparatus, adjusting the frequency, and observing the resulting wave patterns.	[An,S][5]
7	Describe the procedure for demonstrating total internal reflection using a laser. Include details about the setup, the materials required, the angle of incidence you would use, and how you would visually observe the total internal reflection phenomenon	[An,S][5]
8	Describe the procedure you would follow to determine the height of an object using laser triangulation. Include details about the equipment needed, the setup configuration, and how you would calculate the height based on the angles and distances measured.	[An,S][5]
9	Demonstrate the refraction of light through a prism. Describe the setup you would use, including the arrangement of the light source, prism, and screen. Explain how you would measure the angles of incidence and refraction, and how you would observe and illustrate the resulting spectrum of colors formed.	[An,S][5]